Skills Review

Exercise No. 14

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Course/Section: CPE4B **Date: Oct. 7, 2026**

Machine Problem:

Create a program using Interrupt Function 10h that will display the mobile phone below with your desired colors and coordinates. Use string character printing and cursor positioning to display whatever information you want to be displayed in the LCD of the mobile phone. Graph the coordinates on the figure below. Save the program using Exer14.asm as the filename.

Program:

```
; AARON LUDWIG A. ALTAR
; CPE4B
; Exer14 - Mobile Phone (Interrupt 10h)

.model small
.stack
.data

var1    DB 'V   13:26    #','$'
var2    DB 'V   iGlobe   #','$'
var3    DB 'V            #','$'
var4    DB 'V  AARON L.  #','$'
var5    DB 'O    CPE4B   #','$'
var6    DB 'Menu     Names','$'
var7    DB 'CALL   OK   END','$'
var8    DB '1     2     3','$'
var9    DB '4     5     6','$'
var10   DB '7     8     9','$'
var11   DB '*     0     #','$'

.code
start:

        ; screen background (0,0 - 24,79)
        mov ah, 06h             ; Function 06h - scroll up window
        mov bh, 1Fh             ; blue background, white text
        mov ch, 0               ; start row
        mov cl, 0               ; start column
```

```
        mov dh, 24              ; end row
        mov dl, 79              ; end column
        int 10h                 ; BIOS video interrupt - display the window

        ; antenna (0,32 - 2,33)
        mov ah, 06h
        mov bh, 00h             ; black
        mov ch, 0
        mov cl, 32
        mov dh, 2
        mov dl, 33
        int 10h

        ; phone body (3,30 - 24,49)
        mov ah, 06h
        mov bh, 70h             ; white
        mov ch, 3
        mov cl, 30
        mov dh, 24
        mov dl, 49
        int 10h

        ; earpiece (4,37 - 4,42)
        mov ah, 06h
        mov bh, 00h             ; black
        mov ch, 4
        mov cl, 37
        mov dh, 4
        mov dl, 42
        int 10h

        ; LCD bezel (5,32 - 12,47)
        mov ah, 06h
        mov bh, 00h             ; black
        mov ch, 5
        mov cl, 32
        mov dh, 12
        mov dl, 47
        int 10h
```

```
        ; LCD screen (6,33 - 11,46)
        mov ah, 06h
        mov bh, 20h             ; green, black text
        mov ch, 6
        mov cl, 33
        mov dh, 11
        mov dl, 46
        int 10h

        ; <---- control keys ---->
        ; call key (14,32 - 14,35)
        mov ah, 06h
        mov bh, 2Fh             ; green, white text
        mov ch, 14
        mov cl, 32
        mov dh, 14
        mov dl, 35
        int 10h

        ; OK / navigation key (14,38 - 14,41)
        mov ah, 06h
        mov bh, 1Fh             ; blue, white text
        mov ch, 14
        mov cl, 38
        mov dh, 14
        mov dl, 41
        int 10h

        ; end key (14,44 - 14,47)
        mov ah, 06h
        mov bh, 4Fh             ; red, white text
        mov ch, 14
        mov cl, 44
        mov dh, 14
        mov dl, 47
        int 10h

        ; <---- keypad (4 rows x 3 keys) ---->
        ; key 1 (16,32 - 16,35)
        mov ah, 06h
```

```
        mov bh, 0Fh             ; black, white text
        mov ch, 16
        mov cl, 32
        mov dh, 16
        mov dl, 35
        int 10h

        ; key 2 (16,38 - 16,41)
        mov ah, 06h
        mov bh, 0Fh             ; black, white text
        mov ch, 16
        mov cl, 38
        mov dh, 16
        mov dl, 41
        int 10h

        ; key 3 (16,44 - 16,47)
        mov ah, 06h
        mov bh, 0Fh             ; black, white text
        mov ch, 16
        mov cl, 44
        mov dh, 16
        mov dl, 47
        int 10h

        ; key 4 (18,32 - 18,35)
        mov ah, 06h
        mov bh, 0Fh             ; black, white text
        mov ch, 18
        mov cl, 32
        mov dh, 18
        mov dl, 35
        int 10h

        ; key 5 (18,38 - 18,41)
        mov ah, 06h
        mov bh, 0Fh             ; black, white text
        mov ch, 18
        mov cl, 38
        mov dh, 18
```

```
        mov dl, 41
        int 10h

        ; key 6 (18,44 - 18,47)
        mov ah, 06h
        mov bh, 0Fh             ; black, white text
        mov ch, 18
        mov cl, 44
        mov dh, 18
        mov dl, 47
        int 10h

        ; key 7 (20,32 - 20,35)
        mov ah, 06h
        mov bh, 0Fh             ; black, white text
        mov ch, 20
        mov cl, 32
        mov dh, 20
        mov dl, 35
        int 10h

        ; key 8 (20,38 - 20,41)
        mov ah, 06h
        mov bh, 0Fh             ; black, white text
        mov ch, 20
        mov cl, 38
        mov dh, 20
        mov dl, 41
        int 10h

        ; key 9 (20,44 - 20,47)
        mov ah, 06h
        mov bh, 0Fh             ; black, white text
        mov ch, 20
        mov cl, 44
        mov dh, 20
        mov dl, 47
        int 10h

        ; key * (22,32 - 22,35)
```

```
        mov ah, 06h
        mov bh, 0Fh             ; black, white text
        mov ch, 22
        mov cl, 32
        mov dh, 22
        mov dl, 35
        int 10h

        ; key 0 (22,38 - 22,41)
        mov ah, 06h
        mov bh, 0Fh             ; black, white text
        mov ch, 22
        mov cl, 38
        mov dh, 22
        mov dl, 41
        int 10h

        ; key # (22,44 - 22,47)
        mov ah, 06h
        mov bh, 0Fh             ; black, white text
        mov ch, 22
        mov cl, 44
        mov dh, 22
        mov dl, 47
        int 10h

        mov ax, @data           ; initialize the data segment
        mov ds, ax

        ; <---- LCD text ---->
        mov ah, 02h             ; Function 02h - set cursor position
        mov bh, 00h             ; page 0
        mov dh, 6               ; row
        mov dl, 33              ; column
        int 10h
        mov ah, 09h             ; Function 09h - display string
        mov dx, offset var1     ; signal / time / battery
        int 21h

        mov ah, 02h
```

```
        mov bh, 00h
        mov dh, 7
        mov dl, 33
        int 10h
        mov ah, 09h
        mov dx, offset var2    ; network name
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 8
        mov dl, 33
        int 10h
        mov ah, 09h
        mov dx, offset var3    ; signal / battery
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 9
        mov dl, 33
        int 10h
        mov ah, 09h
        mov dx, offset var4    ; owner name
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 10
        mov dl, 33
        int 10h
        mov ah, 09h
        mov dx, offset var5    ; signal / section / battery
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 11
        mov dl, 33
        int 10h
```

```
        mov ah, 09h
        mov dx, offset var6    ; soft keys
        int 21h

        ; <---- key labels ---->
        mov ah, 02h
        mov bh, 00h
        mov dh, 14
        mov dl, 32
        int 10h
        mov ah, 09h
        mov dx, offset var7    ; control key labels
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 16
        mov dl, 33
        int 10h
        mov ah, 09h
        mov dx, offset var8    ; keys 1 2 3
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 18
        mov dl, 33
        int 10h
        mov ah, 09h
        mov dx, offset var9    ; keys 4 5 6
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 20
        mov dl, 33
        int 10h
        mov ah, 09h
        mov dx, offset var10   ; keys 7 8 9
        int 21h
```

```
        mov ah, 02h
        mov bh, 00h
        mov dh, 22
        mov dl, 33
        int 10h
        mov ah, 09h
        mov dx, offset var11    ; keys * 0 #
        int 21h

        mov ah, 02h             ; move the cursor near the bottom so the
        mov bh, 00h             ; DOS prompt will not cover the design
        mov dh, 23
        mov dl, 0
        int 10h

        mov ah, 4Ch             ; terminate program
        int 21h

end start
```

Output:

